Lab 2: Document Retrieval using Inverted Files

Information Retrieval (417524B) - SPPU BE AI&DS

Implement a program for retrieval of documents using inverted files.

CODE

```
documents = {
    1: "Information retrieval is the process of retrieving relevant documents",
    2: "An inverted index maps each term to the documents containing it",
    3: "Python is a popular programming language for data science",
    4: "Search engines use inverted indexes for fast retrieval",
    5: "Machine learning models learn patterns from data",
}

inverted = {}
for doc_id, text in documents.items():
    for term in text.lower().split():
        inverted.setdefault(term, set()).add(doc_id)

print("Inverted Index:")
for term in sorted(inverted):
    print(f"  {term}: {sorted(inverted[term])}")

queries = ["inverted index", "data", "retrieval python"]
for q in queries:
    results = set(documents)
    for t in q.lower().split():
        results &= inverted.get(t, set())
    print(f"\nQuery '{q}' -> documents {sorted(results)}")

# queries.append("learning models")
```

OUTPUT

```
Inverted Index:
  a: [3]
  an: [2]
  containing: [2]
  data: [3, 5]
  documents: [1, 2]
  each: [2]
  engines: [4]
  fast: [4]
  for: [3, 4]
  from: [5]
  index: [2]
  indexes: [4]
  information: [1]
  inverted: [2, 4]
  is: [1, 3]
  it: [2]
  language: [3]
  learn: [5]
  learning: [5]
  machine: [5]
  maps: [2]
  models: [5]
  of: [1]
  patterns: [5]
  popular: [3]
  process: [1]
  programming: [3]
  python: [3]
  relevant: [1]
  retrieval: [1, 4]
  retrieving: [1]
  science: [3]
  search: [4]
  term: [2]
  the: [1, 2]
  to: [2]
  use: [4]
```

```
Query 'inverted index' -> documents [2]

Query 'data' -> documents [3, 5]

Query 'retrieval python' -> documents []
```